$$\frac{1-\cos\theta}{\theta} = \left(\sin\frac{\theta}{2}\right) \frac{\sin\left(\frac{\theta}{2}\right)}{\left(\frac{\theta}{2}\right)}$$

$$\text{Therefore, } \lim_{\theta\to 0}\frac{1-\cos\theta}{\theta} = \lim_{\theta\to 0}\left(\sin\frac{\theta}{2}\right).\lim_{\theta\to 0}\left(\frac{\frac{\sin\theta}{2}}{\frac{\theta}{2}}\right)$$

$$= 0 \times 1 = 0.$$ ■

9.2.10 Some important other limits

Result 9.2

$$\lim_{x\to 0}\frac{e^x-1}{x} = 1.$$

Proof

$$e^x = 1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots \qquad \text{(from sequences and series)}$$

$$\lim_{x\to 0}\frac{e^x-1}{x} = \lim_{x\to 0}\left(\frac{\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots}{x}\right)$$

$$= \lim_{x\to 0}\left(1+\frac{x}{2!}+\frac{x^2}{3!}+\ldots\right) = 1.$$ ■

Result 9.3

$$\lim_{x\to 0}\frac{a^x-1}{x} = \log a,\ a>0$$

Proof

We know that $a^x, \log_a x$ are inverses of each other.

Since $\log f(x)$ is the inverse of $\exp(f(x))$, $\exp(\log f(x)) = f(x)$.

Therefore, $a^x = \exp(\log a^x)$

$$= e^{x\log a}$$

$$\text{Therefore, } \frac{a^x-1}{x} = \frac{e^{x\log a}-1}{x\log a}\times\log a$$

$$\text{Now as } x\to 0,\ y = x\log a \to 0$$

$$\text{Therefore, } \lim_{x\to 0}\frac{a^x-1}{x} = \lim_{y\to 0}\frac{e^y-1}{y}\times\log a = \log a\lim_{y\to 0}\left(\frac{e^y-1}{y}\right) = \log a$$ ■

$$\left(\text{since } \lim_{x\to 0}\frac{e^x-1}{x} = 1\right).$$

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